

6 > Introduction to Quantum Computing

2019, where their quantum computer performed a calculation infeasible for classical computers.

Basic Principles of Quantum Mechanics

Welcome to the bizarre and fascinating realm of quantum mechanics! Buckle up as we explore some of its most mind-bending concepts.

Wave-Particle Duality: Imagine a world where light can act like a wave, diffracting around objects and interfering with itself, but also like a particle, delivering localized packets of energy called photons. That's wave-particle duality! It states that quantum entities like electrons and photons possess both wave and particle-like properties, depending on the experiment. Light's wave nature explains its diffraction and interference patterns, while its particle nature explains the photoelectric effect. This duality challenges our classical understanding of the world, where things are either waves or particles, not both.

Superposition: Now, imagine an electron existing in multiple states simultaneously! That's superposition, another cornerstone of quantum mechanics. Before measurement, a quantum system (like an electron) can exist in a superposition of all possible states, described by a wave function. This "cloud" of possibilities represents the system's probabilistic existence. Only when measured does the wave function collapse, and the system assumes a definite state. It's like a coin spinning in the air; it's neither heads nor tails until it lands.

Uncertainty Principle: But measuring these quantum entities isn't as simple as it seems. Heisenberg's uncertainty principle throws in a wrench. It states that you cannot precisely know both the position and momentum of a particle simultane-

ously. The more accurately you measure one, the less precise your knowledge of the other becomes. This inherent fuzziness arises from the interaction between measurement and the quantum system itself. Imagine trying to measure the position of a butterfly with a net; the act of catching it alters its momentum, blurring the measurement.

Entanglement: Now, let's get truly mind-bending: entanglement. Imagine two particles, created together, becoming "linked" in a way that defies classical physics. If you measure the state of one entangled particle, instantaneously, regardless of distance, the other particle takes the opposite state! This spooky action at a distance has no classical explanation but is crucial for quantum computing. Entangled particles act as a single quantum system, even when separated by vast distances, enabling complex calculations impossible with classical computers.

These are just tiny glimpses into the bizarre and fascinating world of quantum mechanics. These concepts may seem counterintuitive, but they are well-established by experiments and form the foundation of technologies like quantum computers and cryptography.

Quantum Computing vs. Classical Computing

Bits vs. Qubits. That's what it all comes down to in the end. You see, classical computing is built on the foundation of bits, the smallest unit of data, represented by 0s and 1s. This binary system is the cornerstone of all classical digital computing and information processing, enabling complex calculations through sequences of bits in well-defined states. In contrast, quantum computing introduces the qubit, or quantum bit, which rep-

